

Lecture 08B

Eigenvectors and Eigenvalues
Eigenspace and Characteristic Polynomial

Consider the following matrix

$$A = \begin{bmatrix} 0.2 & 0.3 & 0.5 \\ 0.1 & 0.4 & 0.5 \\ 0.6 & 0.2 & 0.2 \end{bmatrix} \text{ and let } v = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Verify that v is an eigenvector of A .

What is its corresponding eigenvalue?

- ✓ 1. $\lambda = 1$
2. $\lambda = 2$
3. $\lambda = 3$

Recall:

$$A v = \lambda v$$

- $\lambda \rightarrow$ eigen value
- $v \rightarrow$ eigen vector

$$A v = \begin{bmatrix} 0.2 & 0.3 & 0.5 \\ 0.1 & 0.4 & 0.5 \\ 0.6 & 0.2 & 0.2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= 1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$\downarrow$
eigen value

Consider the following matrix

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0 \times \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

ignore $\leftarrow \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \neq \lambda \begin{bmatrix} 0 \\ 1 \end{bmatrix}$
for all $\lambda \in \mathbb{R}$

Which of the following is an eigenvalue-eigenvector pair?

1. $v = (0,1)^T$ with corresponding $\lambda = 1$.
2. $v = (1,0)^T$ with corresponding $\lambda = 1$.
3. $v = (0,1)^T$ with corresponding $\lambda = 0$.
- ✓ 4. $v = (1,0)^T$ with corresponding $\lambda = 0$.

Lecture 08 – Eigenvectors + Eigenvalues

- Finding Eigenvalues and Eigenvectors
 - Characteristic Polynomial $\leadsto$ eigen value
 - Eigenspace = $\{ \text{set of all eigenvectors} \} \cup \{ \mathbf{0} \}$

Today, we compute...

Matrix	Characteristic Polynomial	Eigenvalues	Eigenvectors / Eigenspaces
$\begin{bmatrix} 3 & 2 \\ -1 & 6 \end{bmatrix}$			
$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$			
$\begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}$			
$\begin{bmatrix} 0.70 & 0.45 \\ 0.30 & 0.55 \end{bmatrix}$			
$\begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix}$			

Example. Consider the following matrix

$$A = \begin{bmatrix} 0.2 & 0.3 & 0.5 \\ 0.1 & 0.4 & 0.5 \\ 0.6 & 0.2 & 0.2 \end{bmatrix} \text{ and let } v = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Verify that v is an eigenvector of A .
What is its corresponding eigenvalue?

$\hat{A}$ is a square matrix

Definition. (λ, v) occurs as a PAIR
aka eigenvalue-eigenvector pair
Let A be an $n \times n$ -matrix.

A nonzero $v \in \mathbb{R}^n$ is an
eigenvector of A if

$$Av = \lambda v \text{ for some } \lambda \in \mathbb{R}.$$

Here, we say λ is the eigenvalue
corresponding to eigenvector v .

*eigenvectors are non-zero
- λ may be zero

For this question, to verify that v is an eigenvector
we simply compute $\hat{A}v$

$\rightarrow \hat{A}v = \lambda v$ for some λ , then good!

$\rightarrow \hat{A}v \neq \lambda v$ for all λ , then v is NOT an eigenvector.

Real Question: To find λ and v , without any
candidates

Example. Consider the following matrix

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

What is a possible eigenvalue-eigenvector pair?

Consider $\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \underline{v}$,

$$\text{Then } A\underline{v} = \underline{0} = 0 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Here, $\underline{v}$ is eigenvector with eigenvalue 0.

Now, $A\underline{v} = \underline{0}$ means that $\underline{v}$ belongs to $\text{null}(A)$.

$\Rightarrow A$ is NOT invertible.

Definition.

Let A be an $n \times n$ -matrix.

A nonzero $\underline{v} \in \mathbb{R}^n$ is an eigenvector of A if

$$A\underline{v} = \lambda \underline{v} \text{ for some } \lambda \in \mathbb{R}.$$

Here, we say λ is the eigenvalue corresponding to eigenvector $\underline{v}$.

Prop $\lambda = 0$ is an eigenvalue of A

$\Leftrightarrow \text{null}(A) \neq \{0\}$ i.e. $\text{null}(A)$ is non trivial

$\Leftrightarrow A$ is NOT invertible

Example. Consider the matrix

$$A = \begin{bmatrix} 3 & 2 \\ -1 & 6 \end{bmatrix}.$$

What are its eigenvalues and its corresponding eigenvectors?

$$\begin{bmatrix} 3 & 2 \\ -1 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \lambda x \\ \lambda y \end{bmatrix}$$

→ This is NOT a linear system in unknowns λ, x, y

Step 1: Guess the λ

s.t. "the nonlinear system becomes linear",

Step 2: Solve for x and y .

Definition.

Let A be an $n \times n$ -matrix.

A nonzero $\mathbf{v} \in \mathbb{R}^n$ is an eigenvector of A if

$$A\mathbf{v} = \lambda\mathbf{v} \text{ for some } \lambda \in \mathbb{R}.$$

Here, we say λ is the eigenvalue corresponding to eigenvector $\mathbf{v}$.

Example. Consider the matrix

$$A = \begin{bmatrix} 3 & 2 \\ -1 & 6 \end{bmatrix}.$$

What are its eigenvalues and its corresponding eigenvectors?

$$\begin{bmatrix} 3 & 2 \\ -1 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\Rightarrow \left(\begin{bmatrix} 3 & 2 \\ -1 & 6 \end{bmatrix} - \lambda \right) \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

need matrix λ to be able to subtract

$$\Rightarrow \left(\begin{bmatrix} 3 & 2 \\ -1 & 6 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right) \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3-\lambda & 2 \\ -1 & 6-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

* Remember $\begin{bmatrix} x \\ y \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 3-\lambda & 2 \\ -1 & 6-\lambda \end{bmatrix}$ to be NOT invertible

Definition.

Let A be an $n \times n$ -matrix.

A nonzero $\mathbf{v} \in \mathbb{R}^n$ is an eigenvector of A if

$$A\mathbf{v} = \lambda\mathbf{v} \text{ for some } \lambda \in \mathbb{R}.$$

Here, we say λ is the eigenvalue corresponding to eigenvector $\mathbf{v}$.

Example. Consider the matrix

$$A = \begin{bmatrix} 3 & 2 \\ -1 & 6 \end{bmatrix}.$$

What are its eigenvalues and its corresponding eigenvectors?

Definition.

Let A be an $n \times n$ -matrix.

A nonzero $\mathbf{v} \in \mathbb{R}^n$ is an eigenvector of A if

$$A\mathbf{v} = \lambda\mathbf{v} \text{ for some } \lambda \in \mathbb{R}.$$

Here, we say λ is the eigenvalue corresponding to eigenvector $\mathbf{v}$.

$\begin{bmatrix} 3-\lambda & 2 \\ -1 & 6-\lambda \end{bmatrix}$ is NOT invertible

$\Rightarrow$ det is zero

$$\Rightarrow (3-\lambda)(6-\lambda) - (-1)(2) = 0$$

$$\Rightarrow 18 - 9\lambda + \lambda^2 + 2 = 0$$

$$\Rightarrow 20 - 9\lambda + \lambda^2 = 0$$

$$\Rightarrow (5-\lambda)(4-\lambda) = 0$$

$$\therefore \lambda = 5 \text{ and } \lambda = 4$$

$\therefore 5$ & 4 are eigenvalues of A

Example. Consider the matrix

$$A = \begin{bmatrix} 3 & 2 \\ -1 & 6 \end{bmatrix}.$$

What are its eigenvalues and its corresponding eigenvectors?

$$\begin{bmatrix} 3 & 2 \\ -1 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 5 \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\Rightarrow \left(\begin{bmatrix} 3 & 2 \\ -1 & 6 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} \right) \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -2 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \left\{ x = s, y = s : s \in \mathbb{R} \right\}$$

Pick $s \neq 0$. So, an eigenvector is $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ with eigenvalue 5.

$$\text{Set } \lambda = 5$$

Definition.

Let A be an $n \times n$ -matrix.

A nonzero $v \in \mathbb{R}^n$ is an eigenvector of A if

$$Av = \lambda v \text{ for some } \lambda \in \mathbb{R}.$$

Here, we say λ is the eigenvalue corresponding to eigenvector v .

Example. Consider the matrix

$$A = \begin{bmatrix} 3 & 2 \\ -1 & 6 \end{bmatrix}.$$

What are its eigenvalues and its corresponding eigenvectors?

Definition.

Let A be an $n \times n$ -matrix.

A nonzero $\mathbf{v} \in \mathbb{R}^n$ is an eigenvector of A if

$$A\mathbf{v} = \lambda\mathbf{v} \text{ for some } \lambda \in \mathbb{R}.$$

Here, we say λ is the eigenvalue corresponding to eigenvector $\mathbf{v}$.

$$\begin{bmatrix} 3 & 2 \\ -1 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 4 \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\Rightarrow \left(\begin{bmatrix} 3 & 2 \\ -1 & 6 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \right) \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -1 & 2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \left\{ x = s, y = \frac{1}{2}s : s \in \mathbb{R} \right\}$$

Pick $s \neq 0$. So, an eigenvector is $\begin{bmatrix} 1 \\ 1/2 \end{bmatrix}$ with eigenvalue 4.

$$\text{Set } \lambda = 4$$

Power Iteration

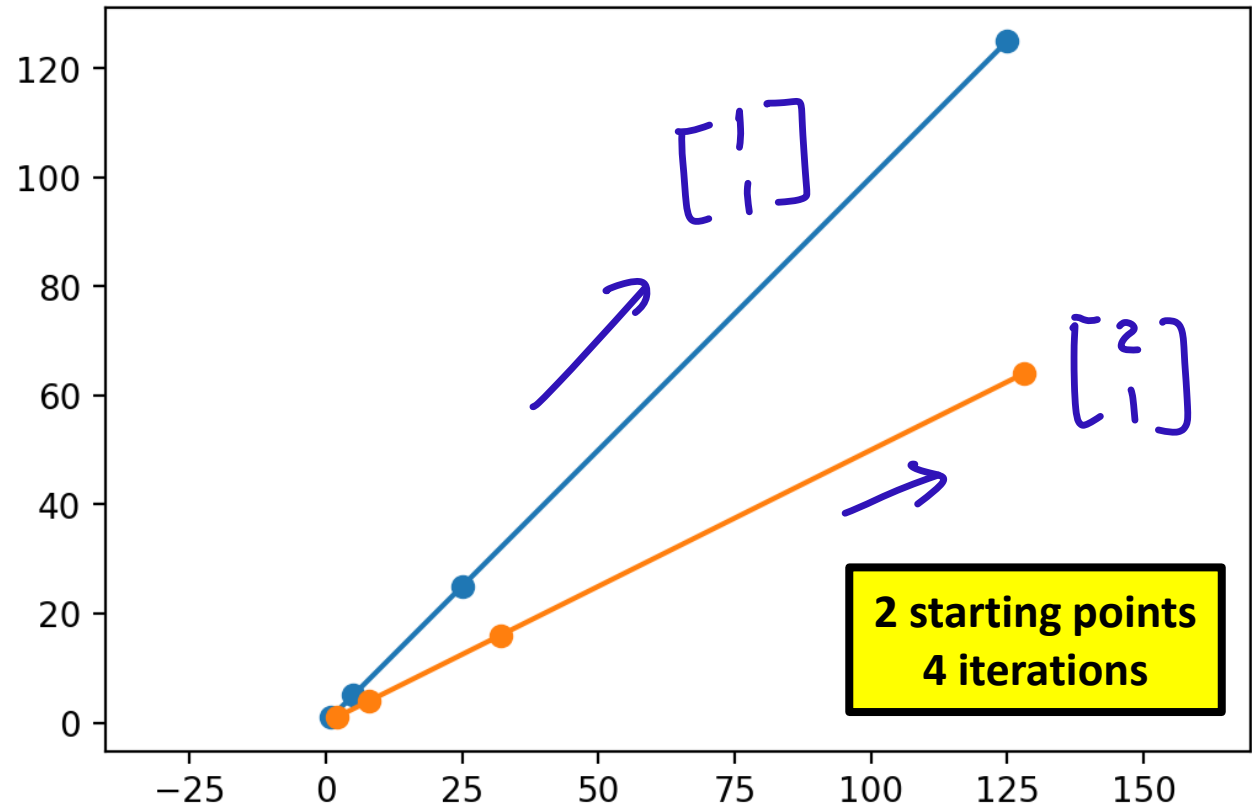
- Let $A = \begin{bmatrix} 3 & 2 \\ -1 & 6 \end{bmatrix}$.
- Start with any point $\begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$.
- Let us compute

$$\begin{bmatrix} x_0 \\ y_0 \end{bmatrix}, A \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}, A^2 \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}, \dots$$

Example. Consider $A = \begin{bmatrix} 3 & 2 \\ -1 & 6 \end{bmatrix}$.

$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector of A with eigenvalue 5.

$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ is an eigenvector of A with eigenvalue 4.



*Indeed, eigen vectors give
"stable directions" (NOT really)

Power Iteration

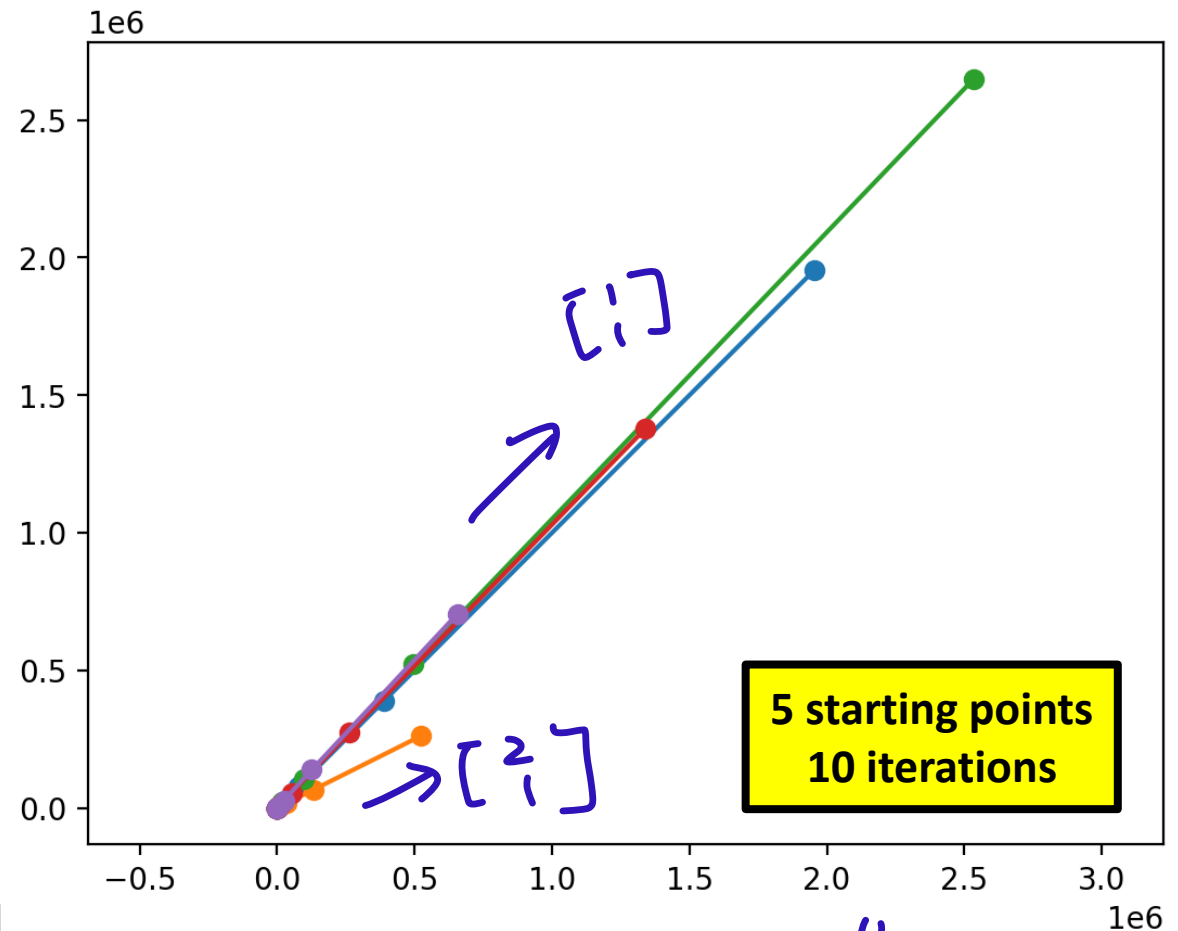
- Let $A = \begin{bmatrix} 3 & 2 \\ -1 & 6 \end{bmatrix}$.
- Start with any point $\begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$.
- Let us compute

$$\begin{bmatrix} x_0 \\ y_0 \end{bmatrix}, A \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}, A^2 \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}, \dots$$

Example. Consider $A = \begin{bmatrix} 3 & 2 \\ -1 & 6 \end{bmatrix}$.

$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector of A with eigenvalue 5.

$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ is an eigenvector of A with eigenvalue 4.



- There is "dominant eigenvector"
i.e almost all starting points will go in
this direction

(TA;PI)

Finding Eigenvalues and Eigenvectors

Let A be an $n \times n$ -matrix.

(Step 1) Find eigenvalues by finding all roots of

$$\det(A - \lambda I). \quad [\text{Solve } \det(\hat{A} - \lambda \hat{I}) = 0]$$

(Step 2) For each eigenvalue λ , determine

$$E_\lambda = \text{null}(A - \lambda I). \quad [\text{Solve } (\hat{A} - \lambda \hat{I}) \underline{v} = \underline{0}]$$

Any nonzero vector in E_λ is an eigenvector corresponding to λ .

Definition

- $\det(\hat{A} - \lambda \hat{I})$
is called the
characteristic poly
of $\hat{A}$.

- E_λ is called the
eigenspace of A
corresponding to λ .

Example. Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}.$$

What are its eigenvalues and its corresponding eigenvectors?

(Step 1) Find eigenvalues by finding all roots of its **characteristic polynomial**

$$\det(A - \lambda I).$$

(Step 2) For each eigenvalue λ , determine the corresponding **eigenspace**.

$$E_\lambda = \text{null}(A - \lambda I).$$

Any nonzero vector in E_λ is an eigenvector corresponding to λ .

Characteristic Poly. $\det(A - \lambda I)$

$$\det \begin{bmatrix} 1-\lambda & 2 & 3 \\ 2 & 4-\lambda & 6 \\ 3 & 6 & 9-\lambda \end{bmatrix}$$

$$= (1-\lambda) \det \begin{pmatrix} 4-\lambda & 6 \\ 6 & 9-\lambda \end{pmatrix} - 2 \det \begin{pmatrix} 2 & 6 \\ 3 & 9-\lambda \end{pmatrix} + 3 \det \begin{pmatrix} 2 & 4-\lambda \\ 3 & 6 \end{pmatrix}$$

co factor expansion

$= \dots$

$$= 14\lambda^2 - \lambda^3 \quad \left(\begin{array}{l} \text{Proof used} \\ \text{sympy} \end{array} \right)$$

$$\Rightarrow \lambda^2(14 - \lambda) = 0$$

Eigenvalues are
 $\lambda = 0$ and $\lambda = 14$ TBEU
 $\downarrow$ multiplicity 2 $\downarrow$ multiplicity 1

Example. Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}.$$

What are its eigenvalues and its corresponding eigenvectors?

(Step 1) Find eigenvalues by finding all roots of its **characteristic polynomial**

$$\det(A - \lambda I).$$

(Step 2) For each eigenvalue λ , determine the corresponding **eigenspace**.

$$E_\lambda = \text{null}(A - \lambda I).$$

Any nonzero vector in E_λ is an eigenvector corresponding to λ .

$$\lambda = 14$$

$$\text{Find } E_{14} = \text{null}(A - 14I)$$

$$\text{Solve } \begin{bmatrix} -13 & 2 & 3 \\ 2 & -10 & 6 \\ 3 & 6 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{matrix} \text{ref} \\ \begin{bmatrix} 1 & 0 & -11/3 \\ 0 & 1 & -2/3 \\ 0 & 0 & 0 \end{bmatrix} \\ \frac{1}{3}t \quad \frac{2}{3}t \quad t \end{matrix}$$

Set of solutions E_{14}

$$= \left\{ x = \frac{1}{3}t, y = \frac{2}{3}t, z = t : t \in \mathbb{R} \right\}$$

One possible eigenvector is $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ (set $t=3$)

Example. Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}.$$

What are its eigenvalues and its corresponding eigenvectors?

(Step 1) Find eigenvalues by finding all roots of its **characteristic polynomial**

$$\det(A - \lambda I).$$

(Step 2) For each eigenvalue λ , determine the corresponding **eigenspace**.

$$E_\lambda = \text{null}(A - \lambda I).$$

Any nonzero vector in E_λ is an eigenvector corresponding to λ .

$$\lambda = 0$$

$$\text{Find } E_0 = \text{null}(\hat{A} - 0 \cdot \hat{I})$$

$$\text{Solve } \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{rref} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$s \quad t$

The set of solutions

$$E_0 = \{ x = -2s - 3t, \quad y = s, \quad z = t : s, t \in \mathbb{R} \}$$

Two linearly independent eigenvectors $\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$

$s=1, t=0 \quad s=0, t=1$

Example. Consider the matrix

$$A = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}.$$

What are its eigenvalues and its corresponding eigenvectors?

(Step 1) Find eigenvalues by finding all roots of its **characteristic polynomial**

$$\det(A - \lambda I).$$

(Step 2) For each eigenvalue λ , determine the corresponding **eigenspace**.

$$E_\lambda = \text{null}(A - \lambda I).$$

Any nonzero vector in E_λ is an eigenvector corresponding to λ .

What is the characteristic polynomial of A ?

$$\det \begin{bmatrix} 0.8 - \lambda & 0.3 \\ 0.2 & 0.7 - \lambda \end{bmatrix}$$

$$= (0.8 - \lambda)(0.7 - \lambda) - (0.2)(0.3)$$

$$= \dots$$

$$= 0.5 - 1.5\lambda + \lambda^2$$

$$= (1 - \lambda)(0.5 - \lambda)$$

Eigenvalues are:

$$\lambda = 1$$

$$\lambda = 0.5$$

Example. Consider the matrix

$$A = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}.$$

What are its eigenvalues and its corresponding eigenvectors?

(Step 1) Find eigenvalues by finding all roots of its **characteristic polynomial**

$$\det(A - \lambda I).$$

(Step 2) For each eigenvalue λ , determine the corresponding **eigenspace**.

$$E_\lambda = \text{null}(A - \lambda I).$$

Any nonzero vector in E_λ is an eigenvector corresponding to λ .

Let $\lambda = 1$.

What is E_1 ?

~~$\text{null}(A)$~~

$\hat{A} - \lambda \hat{I}$ when $\lambda = 1$

$\checkmark \text{null}(\hat{A} - \hat{I})$

solve $\begin{bmatrix} -0.2 & 0.3 \\ 0.2 & -0.3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

set of solutions E_1

$$= \{ x = 3s, y = 2s : s \in \mathbb{R} \}$$

One eigenvector is $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$

Example. Consider the matrix

$$A = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}.$$

What are its eigenvalues and its corresponding eigenvectors?

(Step 1) Find eigenvalues by finding all roots of its **characteristic polynomial**

$$\det(A - \lambda I).$$

(Step 2) For each eigenvalue λ , determine the corresponding **eigenspace**.

$$E_\lambda = \text{null}(A - \lambda I).$$

Any nonzero vector in E_λ is an eigenvector corresponding to λ .

Let $\lambda = 0.5$.

What is $E_{0.5}$?

$$E_{0.5} = \text{null}(A - 0.5I)$$

$$\text{Solve } \begin{bmatrix} 0.3 & 0.3 \\ 0.2 & 0.2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

set of solutions $E_{0.5}$

$$= \{x=s, y=-s: s \in \mathbb{R}\}$$

$\therefore$ One eigenvector is $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

$$\hat{B} = \hat{A}^2$$

Example. Consider the matrix

$$B = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}^2 = \begin{bmatrix} 0.70 & 0.45 \\ 0.30 & 0.55 \end{bmatrix}.$$

What are its eigenvalues and its corresponding eigenvectors?

(Step 1) Find eigenvalues by finding all roots of its **characteristic polynomial**

$$\det(A - \lambda I).$$

(Step 2) For each eigenvalue λ , determine the corresponding **eigenspace**.

$$E_\lambda = \text{null}(A - \lambda I).$$

Any nonzero vector in E_λ is an eigenvector corresponding to λ .

What is the characteristic polynomial of B ?

$$\hat{B} = \begin{bmatrix} 0.7 & 0.45 \\ 0.3 & 0.55 \end{bmatrix}$$

$$\det \begin{bmatrix} 0.7 - \lambda & 0.45 \\ 0.3 & 0.55 - \lambda \end{bmatrix}$$

$$= (0.7 - \lambda)(0.55 - \lambda) - (0.45)(0.3)$$

$$= \dots$$

$$= 0.25 - 1.25\lambda + \lambda^2$$

$$= (0.25 - \lambda)(1 - \lambda)$$

Eigenvalues of $\hat{B}$

$$\lambda = 1 = 1^2$$

$$\lambda = 0.25 = 0.5^2$$

Eigenvalues of $\hat{A} = 1, 0.5$

*You can proceed as usual...

Example. Consider the matrix

$$B = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}^2 = \begin{bmatrix} 0.70 & 0.45 \\ 0.30 & 0.55 \end{bmatrix}.$$

What are its eigenvalues and its corresponding eigenvectors?

(Step 1) Find eigenvalues by finding all roots of its **characteristic polynomial**

$$\det(A - \lambda I).$$

(Step 2) For each eigenvalue λ , determine the corresponding **eigenspace**.

$$E_\lambda = \text{null}(A - \lambda I).$$

Any nonzero vector in E_λ is an eigenvector corresponding to λ .

But let's guess:

Check that eigenvectors are the same.

$\hat{A}$ has eigenvectors $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$, $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

$$\hat{B} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 0.7 & 0.45 \\ 0.30 & 0.55 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} = 1 \cdot \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$\hat{B} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0.7 & 0.45 \\ 0.30 & 0.55 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0.25 \\ -0.25 \end{bmatrix} = 0.25 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

In general, this is true.

Proposition.

$$\hat{A}v = \lambda v$$

Suppose A has eigenvalue λ with corresponding eigenvector v .

Then λ^m is an eigenvalue of A^m with corresponding eigenvector v .

Proof

$$\begin{aligned}\hat{A}^m v &= \hat{A}^{m-1}(\hat{A}v) = \hat{A}^{m-1}(\lambda v) \\ &= \lambda(\hat{A}^{m-1}v) = \lambda(\hat{A}^{m-2})(\hat{A}v) = \lambda(\hat{A}^{m-2})(\lambda v) \\ &= \lambda^2(\hat{A}^{m-2}v) = \dots = \lambda^m v\end{aligned}$$

$\therefore \lambda^m$ is eigen value with eigenvector v

$$\hat{C} = \hat{A}^T$$

Example. Consider the matrix

$$C = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}^T = \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix}.$$

What are its eigenvalues and its corresponding eigenvectors?

(Step 1) Find eigenvalues by finding all roots of its **characteristic polynomial**

$$\det(A - \lambda I).$$

(Step 2) For each eigenvalue λ , determine the corresponding **eigenspace**.

$$E_\lambda = \text{null}(A - \lambda I).$$

Any nonzero vector in E_λ is an eigenvector corresponding to λ .

What is the characteristic polynomial of C ?

$$\det \begin{bmatrix} 0.8 - \lambda & 0.2 \\ 0.3 & 0.7 - \lambda \end{bmatrix}$$

$$= (0.8 - \lambda)(0.7 - \lambda) - (0.3)(0.2)$$

$$= \dots$$

$$= (1 - \lambda)(0.5 - \lambda)$$

↑ same char. poly of $\hat{A}$

∴ eigenvalues are

$$\lambda = 1$$

$$\lambda = 0.5$$

Proposition. Tutorial 7 Q8 for proof

If A has eigenvalue λ , then A^T has eigenvalue λ .

$\hat{A}$ has eigen vectors

$$\begin{bmatrix} 3 \\ 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

lets guess that eigen vectors are the same

$$A^T \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.6 \\ -0.4 \end{bmatrix} \neq \lambda \begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ for all } \lambda$$

Unfortunately, $\hat{A} \neq \hat{A}^T$ do NOT have

the same eigen vectors

↳ but they do share same eigen values
i think that's what prof said.

Example. Consider the matrix

$$C = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}^T = \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix}.$$

What are its eigenvalues and its corresponding eigenvectors?

(Step 1) Find eigenvalues by finding all roots of its **characteristic polynomial**

$$\det(A - \lambda I).$$

(Step 2) For each eigenvalue λ , determine the corresponding **eigenspace**.

$$E_\lambda = \text{null}(A - \lambda I).$$

Any nonzero vector in E_λ is an eigenvector corresponding to λ .

DIY

$$\lambda = 1$$

$$E_1 = \left\{ s \begin{bmatrix} 1 \\ 1 \end{bmatrix} : s \in \mathbb{R} \right\}$$

$$\lambda = 0.5$$

$$E_{0.5} = \left\{ s \begin{bmatrix} 2 \\ -3 \end{bmatrix} : s \in \mathbb{R} \right\}$$

Good-to-know facts

Proposition.

Suppose A has eigenvalue λ with corresponding eigenvector v .

- A^m has eigenvalue λ^m with corresponding eigenvector v .
- If A is invertible, then A^{-1} has eigenvalue $\frac{1}{\lambda}$ with corresponding eigenvector v .
- A^T has eigenvalue λ . Corresponding eigenvector may or may not be v .

Q8, Q9

Tut 7, Q6+Q7

→ go through proof on Mon

Addendum

TA;PI

A	eigenvalues	trace = sum of diagonal entries <i>also $\lambda_1 + \lambda_2$</i>	determinant <i>$\lambda_1 \times \lambda_2$</i>
$\begin{bmatrix} 3 & 2 \\ -1 & 6 \end{bmatrix}$	$\lambda_1 = 5, \lambda_2 = 4$	9	20
$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$	$\lambda_1 = 14, \lambda_2 = \lambda_3 = 0$	14	0
$\begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}$	$\lambda_1 = 1, \lambda_2 = 0.5$	1.5	0.5
$\begin{bmatrix} 0.70 & 0.45 \\ 0.30 & 0.55 \end{bmatrix}$	$\lambda_1 = 1, \lambda_2 = 0.25$	1.25	0.25
$\begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix}$	$\lambda_1 = 1, \lambda_2 = 0.5$	1.5	0.5

Proposition. (Good-to-know fact)

If A has n eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ (counting multiplicities), then $\text{trace}(A) = \sum_{i=1}^n \lambda_i$ and $\det(A) = \prod_{i=1}^n \lambda_i$.

Proof. Tut 7, ~~Q8~~ *Q10* for $n=2,3$

Tut 7, ~~Q8~~ *Q10*
Warning: a bit
challenging